

Trainings ▾

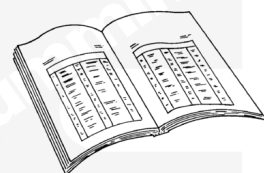
Preparatory Steps

With EGR

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

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ck800wa

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html>)
- Remove the air intake piping.
- Remove the alternator belt. Refer to Procedure 013-005 in Section 13. (</qs3/pubsys2/xml/en/procedures/35/35-013-005-tr.html>)
- Remove the alternator. Refer to Procedure 013-001 in Section 13. (</qs3/pubsys2/xml/en/procedures/35/35-013-001.html>)
- Remove the lubricating oil cooler. Refer to Procedure 007-003 in Section 7. (</qs3/pubsys2/xml/en/procedures/35/35-007-003.html>)
- Remove the coolant filter and filter head. Refer to Procedure 008-007 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-007-tr.html>)
- Remove the coolant heater housing. Refer to Procedure 008-012 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-012-tr.html>)

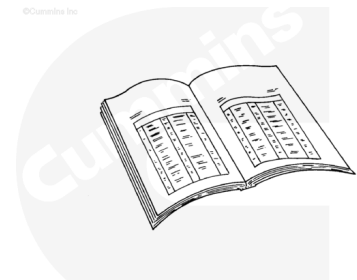
- Remove the turbocharger. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/35/35-010-033-tr.html)
- Remove the coolant thermostat housing and the coolant thermostat housing support. Refer to Procedure 008-015 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-015-tr.html)
- Remove the EGR cooler coolant return line. Refer to Procedure 011-031 in Section 11. (/qs3/pubsys2/xml/en/procedures/35/35-011-031-tr.html)
- Remove the water pump. Refer to Procedure 008-062 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-062-tr.html)

Without EGR

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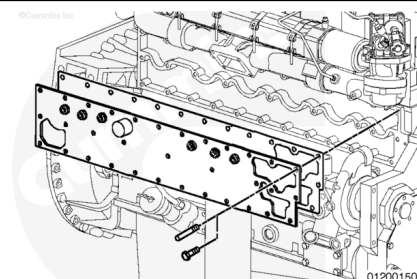
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html)
- Remove the intake air piping.
- Remove the alternator belt. Refer to Procedure 013-005 in Section 13. (/qs3/pubsys2/xml/en/procedures/35/35-013-005-tr.html)
- Remove the alternator. Refer to Procedure 013-001 in Section 13. (/qs3/pubsys2/xml/en/procedures/35/35-013-001-tr.html)
- Remove the lubricating oil cooler. Refer to Procedure 007-003 in Section 7. (/qs3/pubsys2/xml/en/procedures/35/35-007-003-tr.html)

- Remove the coolant filter and filter head. Refer to Procedure 008-007 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-007-tr.html>)
- Remove the coolant heater housing. Refer to Procedure 008-012 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-012-tr.html>)
- Remove the turbocharger. Refer to Procedure 010-033 in Section 10. (</qs3/pubsys2/xml/en/procedures/35/35-010-033-tr.html>)
- Remove the coolant thermostat housing and the coolant thermostat housing support. Refer to Procedure 008-015 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-015-tr.html>)

Remove

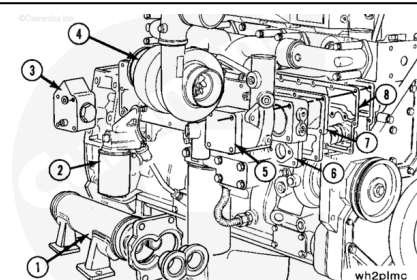
With EGR

Remove the water header plate and gasket.



Without EGR

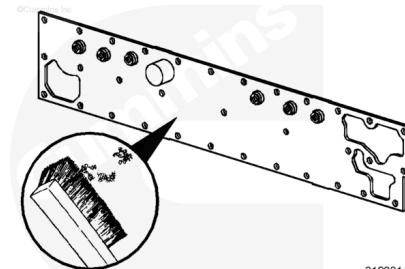
Remove the water header plate (7) and gasket (8).



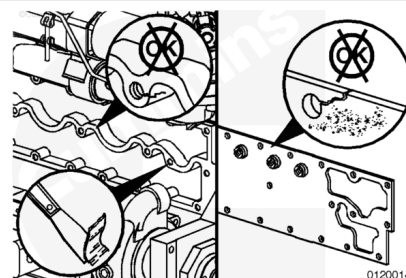
Clean and Inspect for Reuse

With EGR

Use a wire brush and a gasket scraper to clean the water header plate and the cylinder block gasket surface.

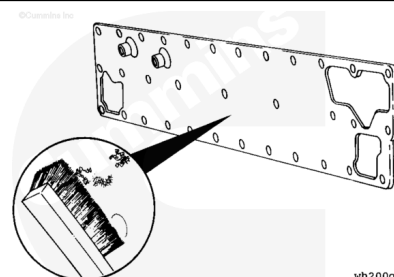


Inspect the water header plate and the cylinder block gasket surface for cracks or other damage.

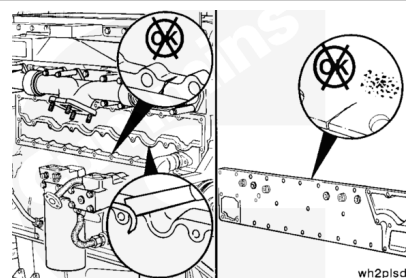


Without EGR

Use a wire brush and a gasket scraper to clean the water header plate and the cylinder block gasket surface.



Inspect the water header plate and the cylinder block gasket surface for cracks or other damage.



Install

With EGR

Install two guide pins, Part Number 3376488, in the top row of the water header plate mounting capscrew holes.

Install the water header plate and capscrews. Use a new gasket

Remove the two guide pins and install the other two capscrews.

The capscrews are **not** tightened to their final value until all related components are installed.

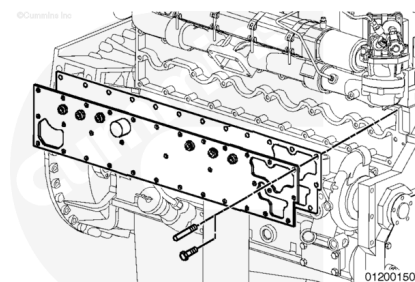
Install the coolant heater housing (3). Refer to Procedure 008-012 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-012-tr.html>)

Install the coolant filter head and filter (2). Refer to Procedure 008-007 in Section 8.

(</qs3/pubsys2/xml/en/procedures/35/35-008-007-tr.html>)

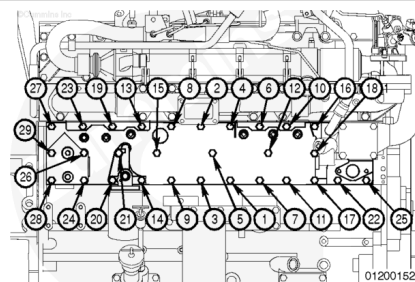
Install the coolant thermostat housing support and gasket (6). Refer to Procedure 008-015 in Section 8.

(</qs3/pubsys2/xml/en/procedures/35/35-008-015-tr.html>)



Tighten all capscrews in the sequence shown.

Torque Value: 54 n•m [40 ft-lb]



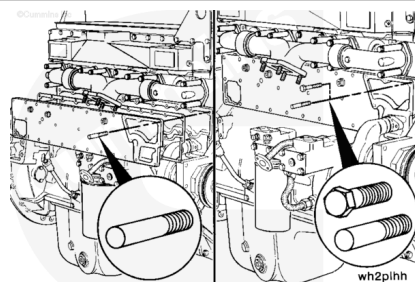
Without EGR

Install two guide pins, Part Number 3376488, in the top row of the water header plate mounting capscrew holes.

Install the water header plate and capscrews. Use a new gasket.

Remove the two guide pins, and install the other two capscrews.

The capscrews are **not** tightened to their final value until all related components are installed.



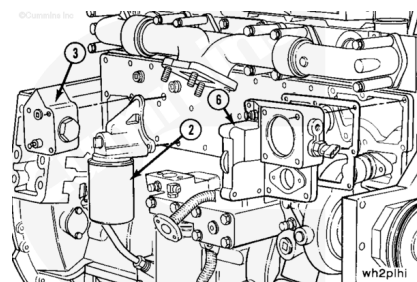
Install the coolant heater housing (3). Refer to Procedure 008-012 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-012-tr.html>)

Install the coolant filter head and filter (2). Refer to Procedure 008-007 in Section 8.

(/qs3/pubsys2/xml/en/procedures/35/35-008-007-tr.html)

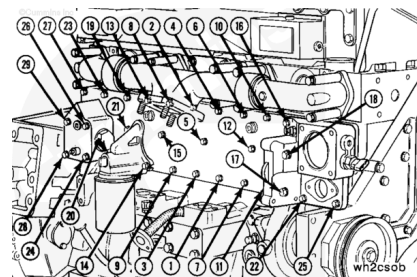
Install the coolant thermostat housing support and gasket (6). Refer to Procedure 008-015 in Section 8.

(/qs3/pubsys2/xml/en/procedures/35/35-008-015-tr.html)



Tighten all capscrews in the sequence shown.

Torque Value: 54 n•m [40 ft-lb]

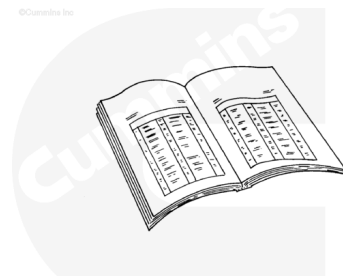


Finishing Steps

With EGR

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⚠ WARNING ⚠

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- Install the water pump. Refer to Procedure 008-062 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-062-tr.html)
- Install the EGR cooler coolant return line. Refer to Procedure 011-031 in Section 11. (/qs3/pubsys2/xml/en/procedures/35/35-011-031-tr.html)
- Install the coolant thermostat housing support and the coolant thermostat housing. Refer to Procedure 008-015

in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-015-tr.html)

- Install the turbocharger. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/35/35-010-033-tr.html)
- Install the coolant heater housing. Refer to Procedure 008-012 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-012-tr.html)
- Install the coolant filter and filter head. Refer to Procedure 008-007 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-007-tr.html)
- Install the lubricating oil cooler. Refer to Procedure 007-003 in Section 7. (/qs3/pubsys2/xml/en/procedures/35/35-007-003-tr.html)
- Install the alternator. Refer to Procedure 007-003 in Section 7. (/qs3/pubsys2/xml/en/procedures/35/35-007-003-tr.html)
- Install the alternator belt. Refer to Procedure 013-005 in Section 13. (/qs3/pubsys2/xml/en/procedures/35/35-013-005-tr.html)
- Install the air intake piping.
- Fill the cooling system. Refer to Procedure 013-001 in Section 13. (/qs3/pubsys2/xml/en/procedures/35/35-013-001-tr.html)
- Operate the engine until it reaches a temperature of 80°C [180°F] and check for coolant leaks.

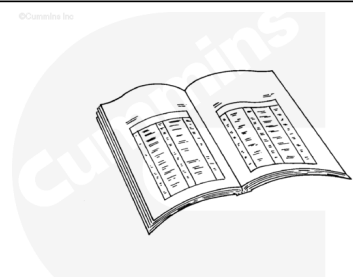
Without EGR

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- Install the coolant thermostat housing support and the coolant thermostat housing. Refer to Procedure 008-015 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-015-tr.html>)
- Install the turbocharger. Refer to Procedure 010-033 in Section 10. (</qs3/pubsys2/xml/en/procedures/35/35-010-033-tr.html>)
- Install the coolant heater housing. Refer to Procedure 008-012 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-012-tr.html>)
- Install the coolant filter and filter head. Refer to Procedure 008-007 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-007-tr.html>)
- Install the lubricating oil cooler. Refer to Procedure 007-003 in Section 7. (</qs3/pubsys2/xml/en/procedures/35/35-007-003-tr.html>)
- Install the alternator. Refer to Procedure 013-001 in Section 13. (</qs3/pubsys2/xml/en/procedures/35/35-013-001-tr.html>)
- Install the alternator belt. Refer to Procedure 013-005 in Section 13. (</qs3/pubsys2/xml/en/procedures/35/35-013-005-tr.html>)
- Install the air intake piping.
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html>)
- Operate the engine until it reaches a temperature of 80°C [180°F] and check for coolant leaks.